Department of Electrical Engineering
विद्युत अभियांत्रिकी विभाग

Electronics Lab
इलेक्ट्रॉनिक्स लैब

Experiment Number 4
प्रयोग संख्या 4

Study of various logic gate IC 7400,7402,7404,7408 and verification of their truth table.

विभिन्न लॉजिक गेट IC 7400,7402,7404,7408 का अध्ययन एवं उनकी सत्यता सारणी का सत्यापन।

MAULANA AZAD NATIONAL INSTITUTEOF TECHNOLOGY, BHOPAL

ELECTRONICS LAB

EXPERIMENT NO.4

Aim:-Study of various logic gate IC 7400, 7402, 7404, 7408, 7432, 7486 and verification of their truth tables.

Instruments required:-

1. Digital IC trainer kit
2. T.T.L IC 7400, 7402, 7404, 7408 & 7432.

Procedure:-

Verification of Truth Table of 'OR' Gate:

1. Connect the A and B inputs of OR Gate to logic inputs. 'O' and '0' as shown in the truth table for OR Gate. Also, connect the output of the OR Gate to the output indicator through patch cords.
2. Switch ON the instrument using OFF/ON toggle switch provided on the front panel.
3. Observe the output indicator. If it glows the indication is that the output is in state '1' and if it does not glow the indication is that the output is in state '0'.
4. Similarly verify the output for other combinations of input 'A' and 'B' as shown in the truth table of OR gate.

Symbol:

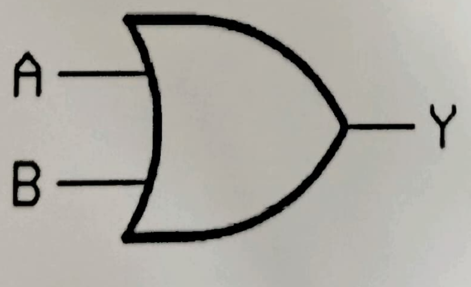

Fig 1.1

Pin Diagram:

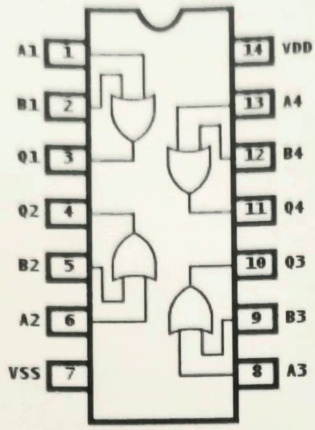

Fig 1.2

Truth Table:

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

Fig 1.3

<u>Verification of Truth Table of 'AND' Gate:</u>

1. Connect the A and B inputs of AND Gate to logic inputs. '0' and '0' as shown in the truth table for AND Gate. Also, connect the output of the AND Gate to the output indicator through patch cords.
2. Switch ON the instrument using OFF/ON toggle switch provided on the front panel.
3. Observe the output indicator. If it glows the indication is that the output is in state '1' and if it does not glow the indication is that the output is in state ‘0’.
4. Similarly verify the output for other combinations of input 'A' and 'B' as shown in the truth table of AND gate.

Symbol:

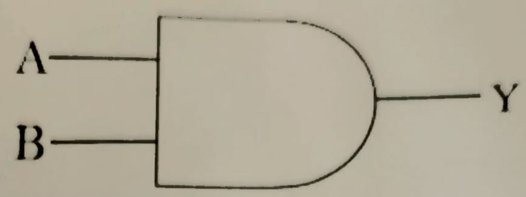

Fig 2.1

Pin Diagram:

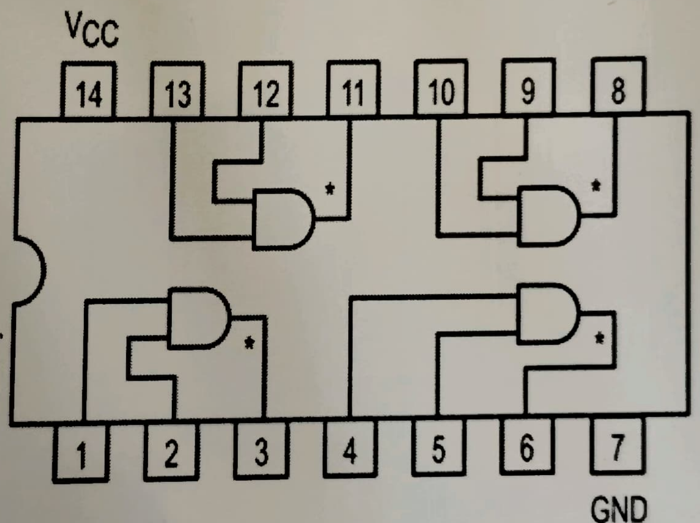

Fig 2.2

Truth table:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Fig 2.3

Verification of Truth Table of 'EX-OR' Gate:

1. Connect the A and B inputs of EX-OR Gate to logic inputs. '0' and '0' as shown in the truth table for EX-OR Gate. Also, connect the output of the EX-OR Gate to the output indicator through patch cords.
2. Switch ON the instrument using OFF/ON toggle switch provided on the front panel.
3. Observe the output indicator. If it glows the indication is that the output is in state '1' and if it does not glow the indication is that the output is in state '0'.
4. Similarly verify the output for other combinations of input 'A' and 'B' as shown in the truth table of EX-OR gate.

Pin Diagram:

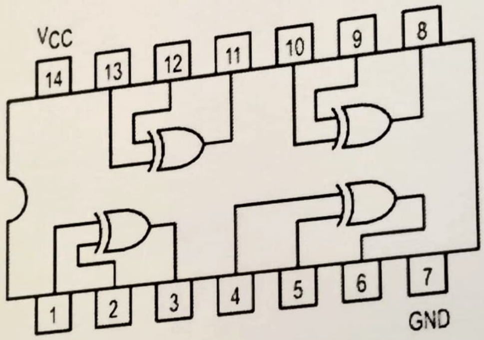

Fig 3.1

Symbol:

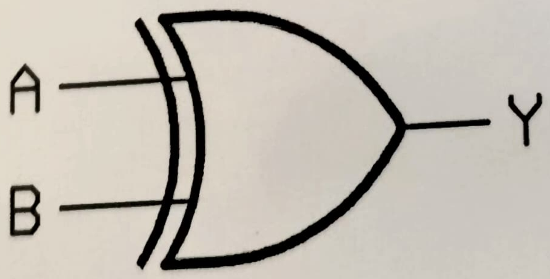

Fig 3.2

Truth table:

Verification of Truth Table of 'NAND' Gate:

1. Connect the A and B inputs of NAND Gate to logic inputs. '0' and '0' as shown in the truth table for NAND Gate. Also, connect the output of the NAND Gate to the output indicator through patch cords.
2. Switch ON the instrument using OFF/ON toggle switch provided on the front panel.
3. Observe the output indicator. If it glows the indication is that the output is in state '1' and if it does not glow the indication is that the output is in state '0'.
4. Similarly verify the output for other combinations of input 'A' and 'B' as shown in the truth table of NAND gate.

Pin Diagram:

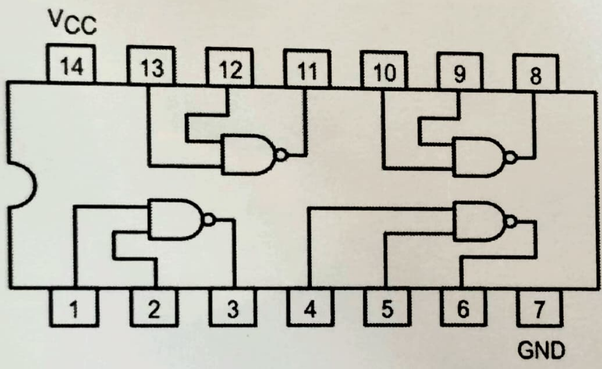

Fig 4.1

Symbol:

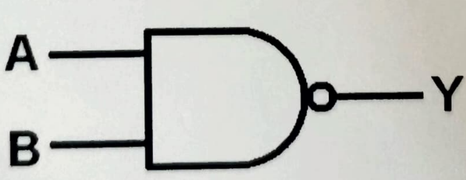

Fig 4.2

Truth table:

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Fig 4.3

Verification of Truth Table of 'NOT' Gate:

5. Connect the A and B inputs of NOT Gate to logic inputs. '0' and '0' as shown in the truth table for NOT Gate. Also, connect the output of the NOT Gate to the output indicator through patch cords.
6. Switch ON the instrument using OFF/ON toggle switch provided on the front panel.
7. Observe the output indicator. If it glows the indication is that the output is in state '1' and if it does not glow the indication is that the output is in state '0'.
8. Similarly verify the output for other combinations of input 'A' and 'B' as shown in the truth table of NOT gate.

Pin Diagram:

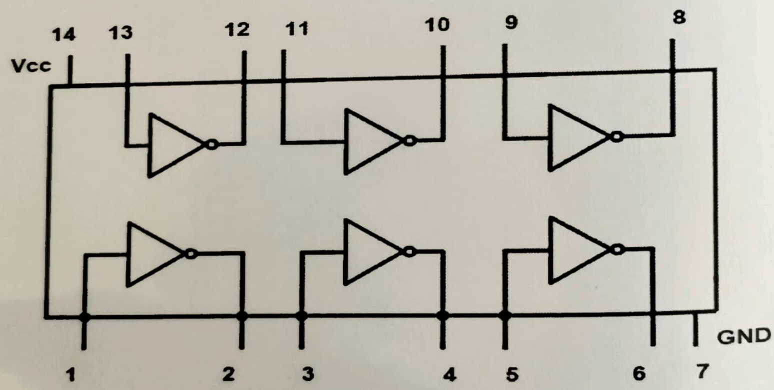

Fig 5.1

Symbol:

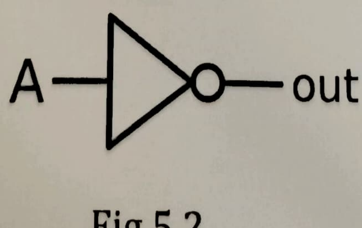

Fig 5.2

Truth table:

A	Y
1	0
0	1

Fig 5.3

Verification of Truth Table of 'NOR' Gate:

1. Connect the A and B inputs of NOR Gate to logic inputs. '0' and '0' as shown in the truth table for NOR Gate. Also, connect the output of the NOR Gate to the output indicator through patch cords.
2. Switch ON the instrument using OFF/ON toggle switch provided on the front panel.
3. Observe the output indicator. If it glows the indication is that the output is in state '1' and if it does not glow the indication is that the output is in state '0'.
4. Similarly verify the output for other combinations of input 'A' and 'B' as shown in the truth table of NOR gate.

Pin Diagram:

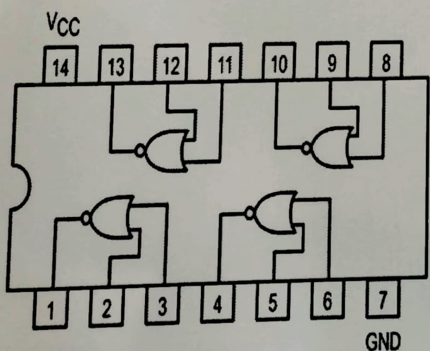

Fig 6.1

Symbol:

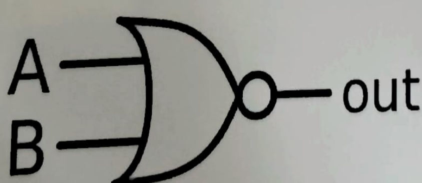

Fig 6.2

Truth table:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Fig 6.3

Verification of Truth Table of 'EX-NOR' Gate:

1. Connect the A and B inputs of the EX- NOR Gate to logic inputs. '0' and '0' as shown in the truth table for EX- NOR Gate. Also, connect the output of the EX-NOR Gate to the output indicator through patch cords.
2. Switch ON the instrument using OFF/ON toggle switch provided on the front panel.
3. Observe the output indicator. If it glows the indication is that the output is in state '1' and if it does not glow the indication is that the output is in state '0'.
4. Similarly verify the output for other combinations of input 'A' and 'B' as shown in the truth table of EX- NOR gate.

Pin Diagram:

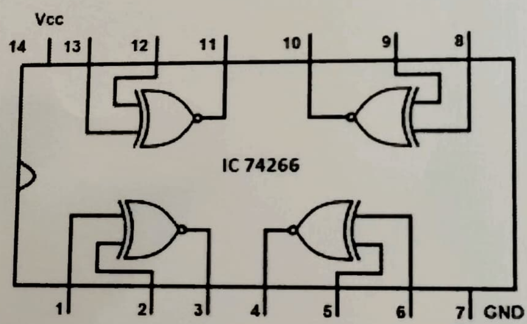

Fig7.1

Symbol:

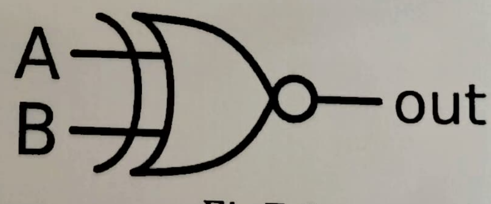

Fig7.2

Truth table:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Fig7.3

Result & Conclusion:

Questions :

1. What is the function of logic gates in digital electronics, and why are they fundamental to circuit design?

2. Explain the concept of a "truth table" in relation to logic gates?

3. What is the difference between the various logic gate ICs (7400, 7402, 7404, 7408, 7432, 7486)?

4. How are logic gates implemented in IC form, and what are the benefits of using logic gate ICs over discrete components?

5. What are the standard package types for these logic gate ICs (7400 series) and their pin configurations?